

Large Language Models, Prompting and AI tools

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01

What Are LLMs?

Understanding how Large Language Models work and why they matter

Topics: Definition · How LLMs Learn · Key Concepts · Capabilities & Limitations

What is a Large Language Model (LLM)?

An LLM is a deep learning model trained on massive amounts of text to understand and generate human language – predicting the most likely next word, one token at a time.

How LLMs Learn

Trained on trillions of words from the internet: books, Wikipedia, code, news, forums
Training adjusts billions of internal 'weights' to learn patterns of language
No explicit rules programmed – patterns emerge from data alone

Parameters & Scale

Parameters are the 'weights' learned during training (GPT-4: ~1 trillion+)
More parameters → more capacity to understand nuance and complex reasoning
Scale is why modern LLMs far outperform older, smaller models

Transformer Architecture

The underlying architecture of all modern LLMs (GPT, Claude, Gemini, Llama)
'Attention mechanism' lets the model focus on relevant parts of long text
Processes entire sequences in parallel – far more powerful than older RNN models

Tokens, Not Words

LLMs process 'tokens' – chunks of text (a word, part of a word, or punctuation)
Context window = how many tokens the model can 'see' at once (up to millions)
Responses are generated token by token – probabilistic, not deterministic

LLM Capabilities & Limitations



What LLMs Do Well

Text generation & summarization

Write, rewrite, summarize, translate – across any style or length

Reasoning & analysis

Break down complex problems, compare options, evaluate arguments

Code writing & debugging

Generate, explain, and fix code across dozens of programming languages

Question answering

Answer factual questions using trained knowledge (with caveats on recency)

Creative tasks

Draft stories, brainstorm ideas, write poetry, roleplay personas



Known Limitations

Hallucination

LLMs can confidently generate false facts, fake citations, or invented details

Knowledge cutoff

Training data has a cutoff date – no awareness of recent events without tools

No true reasoning

Pattern-matching, not logic – complex math or causal reasoning can fail

Context limits

Cannot remember previous conversations (by default) – each session is fresh

Bias & toxicity

Trained on internet data – may reflect human biases present in that data

02

Prompt Engineering

Techniques from promptingguide.ai to get consistently better results from LLMs

Techniques: Zero-Shot · Few-Shot · Chain-of-Thought · RAG · Tree of Thoughts · and more

Prompting Techniques · Foundational Methods



Zero-Shot Prompting

Instruct the model directly with no examples. The model relies entirely on its pretrained knowledge.

When to use

Simple, well-defined tasks where format and intent are clear without demonstration

e.g. 'Classify as positive or negative: I loved the movie.'



Few-Shot Prompting

Provide 2–5 example input/output pairs before the real query to guide format, tone, and style.

When to use

Output format is hard to describe in words; or consistency across responses is critical

e.g. Show 3 labeled examples → ask model to label the 4th.



Chain-of-Thought (CoT)

Ask the model to reason step-by-step before answering. Exposes intermediate reasoning.

When to use

Multi-step math, logic puzzles, causal reasoning, or tasks with sequential dependencies

e.g. 'Think step by step before answering.'



Self-Consistency

Sample multiple independent reasoning paths and pick the most frequent final answer.

When to use

High-stakes reasoning where one sample may be wrong – accuracy matters over speed

e.g. Run CoT 5×, tally the outputs, choose the most common result.



Tree of Thoughts (ToT)

Explore multiple reasoning branches in parallel, evaluate each path, backtrack if needed.

When to use

Open-ended planning, creative problems, or puzzles with no single correct path

e.g. 'Explore three strategies, evaluate each, then choose the best.'



RAG – Retrieval-Augmented Generation

Retrieve relevant documents from a knowledge base and inject them into the prompt.

When to use

Tasks needing up-to-date or domain-specific knowledge beyond the model's training data
e.g. Q&A over internal docs, legal texts, or recent research papers.

Practical Prompt Design · Tips from promptingguide.ai

01

Be specific & detailed

Vague prompts get vague answers. Specify the task, format, length, audience, and tone.

02

Assign a role or persona

'You are an expert software engineer.' Role prompting shifts style and depth of output.

03

Show, don't just tell

Use few-shot examples. Demonstrate the output format you want rather than describing it.

04

Instruct to think first

'Think step by step' or 'think hard before answering' improves reasoning quality.

06

Iterate – don't just send once

First response is a starting point. Give feedback, ask for alternatives, and refine.

07

Avoid leading the model

Don't hint at the answer you want. Neutral framing → honest, unbiased responses.

08

Use delimiters for clarity

Wrap input text in <tags>, triple backticks, or quotes to separate instructions from content.

Specify output format

Ask for JSON, bullet points, a table, or a numbered list. Format constraints reduce post-processing.

SECTION

03

Common LLM Tools

An overview of major AI tools, their strengths, and best use cases

Tools: ChatGPT · Claude · Gemini · Copilot · Perplexity · NotebookLM · Midjourney · Cursor · and more

Common LLM Tools · Conversational AI

ChatGPT (OpenAI)

GPT-4o / o1 / o3

Best for:

General-purpose assistant, coding, writing, image gen, data analysis. Largest user base & plugin ecosystem.

When to use:

Everyday tasks, brainstorming, content creation, coding help.

Claude (Anthropic)

Claude 3.5/4 Sonnet & Opus

Best for:

Long document analysis (200K+ tokens), nuanced writing, reasoning, avoiding sycophancy. Strong safety profile.

When to use:

Large document review, complex reasoning, careful analysis, professional writing.

Gemini (Google)

Gemini 2.0 / Ultra

Best for:

Deep Google integration (Search, Drive, Docs, Gmail). Strong multimodal & real-time web search.

When to use:

Google Workspace tasks, research with real-time data, YouTube/image analysis.

Microsoft Copilot

GPT-4o

Best for:

Embedded in Microsoft 365 (Word, Excel, PowerPoint, Teams, Outlook). Enterprise

When to use:

Office productivity, meeting summaries, Excel analysis,

Tip: These tools share the same core technology. Choose based on ecosystem fit, document size, and specific task – not brand preference.

Common LLM Tools · Specialized & Developer Tools

Research & Search

Perplexity AI

AI search engine that cites sources in real time. Best for: factual research, finding current information with traceable citations.

Research & Docs

NotebookLM

Google's source-grounded AI. Analyzes your uploaded documents. Best for: literature review, study aids, document Q&A.

Image Generation

Midjourney / DALL·E / Stable Diffusion

Text-to-image models. Best for: visual concept generation, design mockups, illustration.

Coding Assistant

GitHub Copilot / Cursor

Code-focused AI embedded in your IDE. Best for: code completion, debugging, code review, refactoring.

Voice & Audio

ElevenLabs / Whisper

Voice cloning and speech-to-text. Best for: podcast production, accessibility features, video narration.

Video Generation

Sora / Kling / Runway

Text-to-video models. Best for: explainer videos, marketing assets, rapid prototyping of visual content.

SECTION

04

NotebookLM Guide

What it is, when to use it, and how to get the most out of it

Topics: What is NotebookLM · When to Use It · Key Features · How to Use Effectively

NotebookLM · What Is It?

NotebookLM is Google's source-grounded AI research assistant (powered by Gemini). Unlike ChatGPT, it ONLY answers from documents YOU upload – not from the internet or general training data. This makes its answers more accurate, traceable, and specific to your materials.

Upload Any Source

PDFs, Google Docs/Slides, YouTube videos, web URLs, .txt files, audio files
Up to 50 sources per notebook (free tier), 500K words per source
AI becomes an expert on your uploaded content – not the broader internet

Studio: Multiple Output Formats

Audio Overview: 2 AI voices discuss your content like a podcast
Mind maps, timelines, slide decks, infographics, quizzes, flashcards
Reports: blog posts, study guides, briefings – all from your sources

Source-Grounded Answers

Every answer includes inline citations linked to exact passages in your docs
Hover over citations to verify – dramatically reduces hallucination
Responses change as you add/remove sources from the notebook

Free & Generous Tier

Free: 100 notebooks, 50 sources each, 50 chat queries/day, 3 audio generations
NotebookLM Plus: 300 sources, higher limits, Deep Research mode
No coding or setup – runs in browser and as a mobile app

NotebookLM · When to Use It

NotebookLM vs. General AI (ChatGPT / Claude):

Task	NotebookLM	General AI
Research from your own documents	✅ Ideal – cites exact passages	⚠️ Makes up plausible-sounding content
Large batch of PDFs (10–50 papers)	✅ Upload all, query across all	⚠️ Must paste one at a time; context fills fast
Avoiding hallucinations	✅ Grounded in your sources only	⚠️ May invent facts confidently
Audio/podcast from documents	✅ One-click Audio Overview	❌ Not a native feature
Real-time web search	❌ No internet access	✅ Web search built-in (Perplexity better)
Memory of past conversations	❌ No memory between sessions	⚠️ Limited, project-based
General writing / creative tasks	⚠️ Limited to uploaded content	✅ Much stronger

NotebookLM · Key Features

Chat with Sources

Ask any question – NotebookLM answers using only your uploaded documents with cited passages you can click to verify.

Audio Overview

Generate a podcast-style discussion between two AI voices summarizing your content. Great for commuting or revision.

Mind Maps & Timelines

Auto-generate visual mind maps and timelines from your sources. Quickly get a structural overview of any complex topic.

Slides & Infographics

Create slide decks, infographics, and visual summaries from your notebooks – all grounded in your uploaded sources.

Flashcards & Quizzes

Instantly generate study aids – flashcards, multiple-choice quizzes – from lecture notes, research papers, or any document.

Save to Notes

Save any AI response to a persistent note linked to its source passage. Organize notes and export to Google Docs.

NotebookLM · How to Use Effectively

Step-by-step workflow:

Create a focused notebook

One notebook per project or topic. A focused set of sources gives sharper, more reliable answers.

02

Upload your sources

PDFs, lecture slides, URLs, YouTube videos. Add the most relevant documents – quality over quantity.

03

Start with suggested questions

NotebookLM auto-suggests questions based on your sources. Use these to quickly orient yourself.

04

Reference specific sources

'Compare Article A and Article B on X topic.' Directing attention to specific sources gives cleaner results.

Save & export insights

Click 'Save to Note' on useful responses. Export notes to Google Docs for collaboration or further editing.



Pro Tips:

Use Audio Overview for complex topics you're unfamiliar with. · Scope queries to subsets of sources for targeted answers. · Create separate notebooks for different course subjects. · Generate a mind map first to understand structure before deep-diving.

NotebookLM · Reading a Textbook Chapter Effectively

Upload the chapter PDF → use NotebookLM as your interactive reading partner – not as a shortcut to skip reading, but as a scaffold that makes every minute of reading count more.

01 Survey before you read

Ask: 'List all headings and subheadings of this chapter and summarize each in one sentence.' This creates a mental map before diving in.

03 Convert headings into questions

Prompt: 'Turn each section heading into a question I should be able to answer after reading.' Use these as your reading goals – purpose-driven reading retains far more than passive reading.

05 Generate flashcards & a study guide

Use Studio → Flashcards to auto-generate cards from the chapter. Use Studio → Study Guide for a one-click structured review. Both are grounded entirely in your uploaded text.

02 Generate a focused summary

Prompt: 'Summarize this chapter in 300 words. Highlight the 3 most important concepts and any key terms I need to know.' Read this before the chapter to know what to look for.

04 Quiz yourself after each section

Prompt: 'Quiz me with 5 questions on [section topic]. Include one where I explain a concept in my own words and one where I apply it to a new scenario. No hints.'

06 Listen via Audio Overview

Generate an Audio Overview of the chapter for a second pass while commuting or exercising. Ask it to 'emphasize key terms and exam-likely concepts' via the Customize option.



Golden rule (Vertech Academy): Use NotebookLM to amplify your reading – not replace it. The AI finds the structure; you do the thinking.